

Lecture 7: Probabilistic Reasoning

DD2421

André Pereira
(slides heavily adapted from Bob Sturm)

My three-lecture block

- Today: probabilistic reasoning
- Next time: learning as inference
- Next next time: priors and latent variables

My three-lecture block

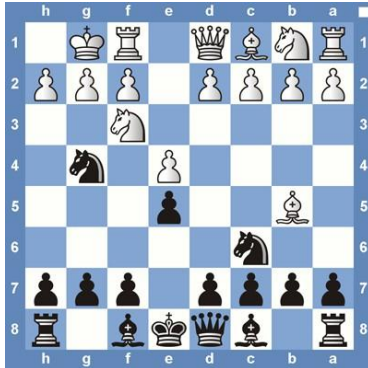
- Today: probabilistic reasoning
- Next time: learning as inference
- Next next time: priors and latent variables

Constructive alignment

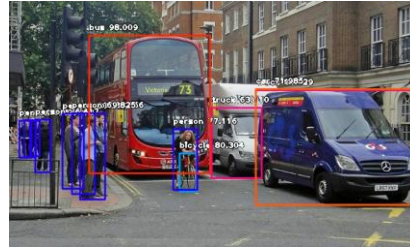
- assignments: part of lab 3 is relevant (Naïve Bayes)
- three exam problems ($\sim 1/3$ of the grade) is on probabilistic methods
- the topic is fundamental for the advanced course (DD2434)

Why Machine Learning?

AI in the 1970s



AI today



We need to deal with uncertainty!

Uncertainty

- How can we deal with the randomness we see around us?
- How can we formally describe degrees of belief?

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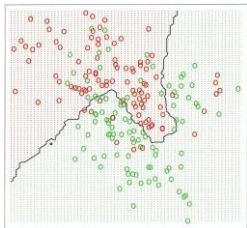
Example: Monty Hall Problem



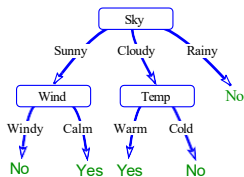
A contestant chooses Door 1. Monty, who knows where the prize is, opens Door 2 and reveals a goat. Monty always opens a door and never opens the door with the prize. If Monty has a choice between two goat doors, he chooses uniformly at random. Should the contestant switch to Door 3?

Examples of ML Methods seen so far

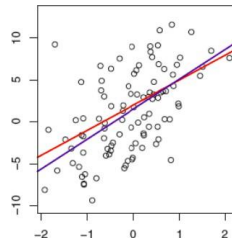
k Nearest Neighbour



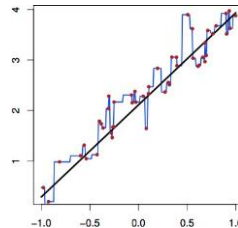
Decision Trees



Least squares Regression



k -NN Regression



Probability Theory in ML

Incorporate probabilistic thinking at all levels

- We start with incomplete knowledge (uncertainty)
- We then reduce uncertainty by incorporating observations
- We then propagate belief and update our estimations.

Probabilistic machine learning provides
a unifying framework for ML!

Probabilistic Methods: Strengths & Challenges

Strengths

- Explicit assumptions
- Quantified uncertainty
- Ability to incorporate prior knowledge
- Modularity/compositionality
- Unified framework for learning and inference

Probabilistic Methods: Strengths & Challenges

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Challenges

- Model assumptions might be wrong
- Exact inference can be intractable
- Approximate inference and learning can be computationally expensive

Outline

1 Probability Theory Reminder

- Axioms and Properties
- Distributions and Moments

2 Probabilistic Machine Learning

- Supervised Learning, General Definition
- Regression
- Classification

Axiomatic definition of probabilities (Kolmogorov)

Given a *sample space* Ω of all possible *outcomes*, and an *event* E , or set of outcomes from Ω (possibly empty), then:

① $P(E) \geq 0$ for all $E \subseteq \Omega$

Axiomatic definition of probabilities (Kolmogorov)

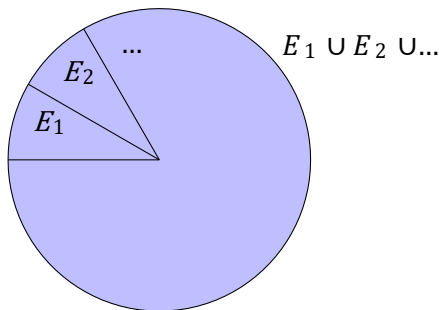
Given a *sample space* Ω of all possible *outcomes*, and an *event* E , or set of outcomes from Ω (possibly empty), then:

- 1 $P(E) \geq 0$ for all $E \subseteq \Omega$
- 2 If $E = \Omega$ then $P(\Omega) = 1$ (sure event) $\Rightarrow 0 \leq P(E) \leq 1$

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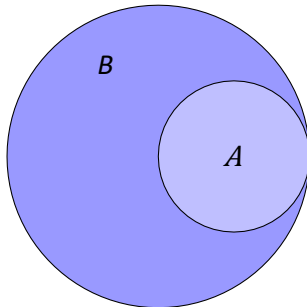
- 1 $P(E) \geq 0$ for all $E \subseteq \Omega$
- 2 If $E = \Omega$ then $P(\Omega) = 1$ (sure event) $\Rightarrow 0 \leq P(E) \leq 1$
- 3 If $E_1, E_2, \dots$ is a countable sequence of pairwise disjoint events, then



$$P(E_1 \cup E_2 \cup \dots) = \sum_i P(E_i)$$

Consequences

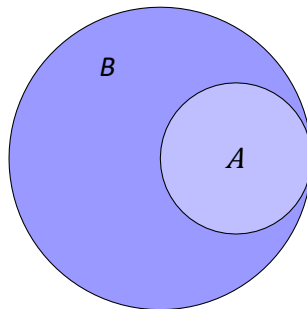
- ① Monotonicity: $P(A) \leq P(B)$ if $A \subseteq B$



Example (die): $A = \{3\}$, $B = \{\text{odd}\}$

Consequences

- 1 Monotonicity: $P(A) \leq P(B)$ if $A \subseteq B$



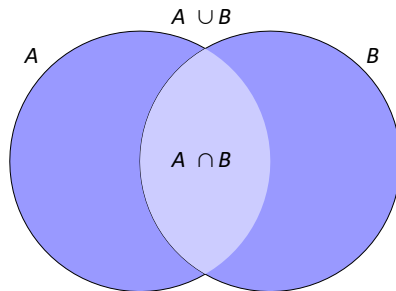
Example (die): $A = \{3\}$, $B = \{\text{odd}\}$

- 2 Empty set $\emptyset$: $P(\emptyset) = 0$

Example (die): $P(A \cap B)$ where $A = \{\text{odd}\}$, $B = \{\text{even}\}$

More Consequences: Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

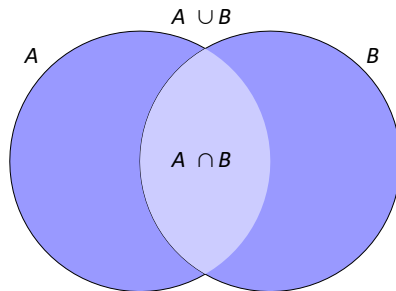


Example:

A	$=$	$\{1, 3, 5\},$	$P(A)$	$=$	$\frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$
B	$=$	$\{5, 6\},$	$P(B)$	$=$	$\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$
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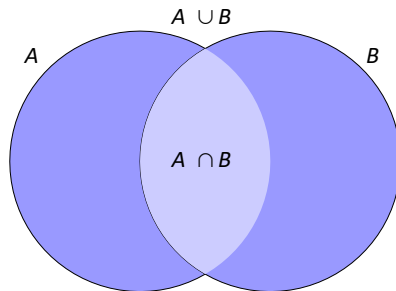


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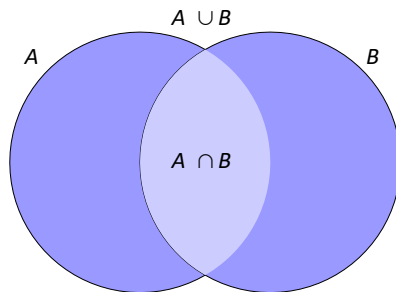


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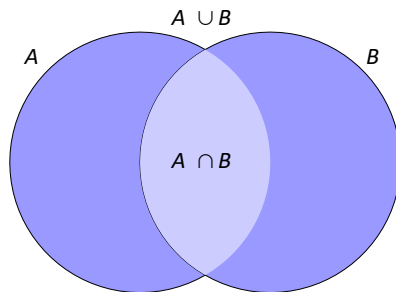


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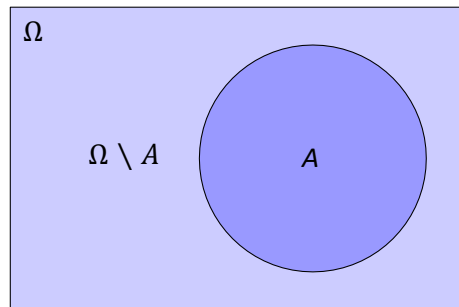


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$A \cup B$	$= \{1, 3, 5, 6\}$	$P(A \cup B)$	$= \frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{2}{3}$

More Consequences: Negation Rule

$$P(\overline{A}) = P(\Omega \setminus A) = 1 - P(A)$$



Example:

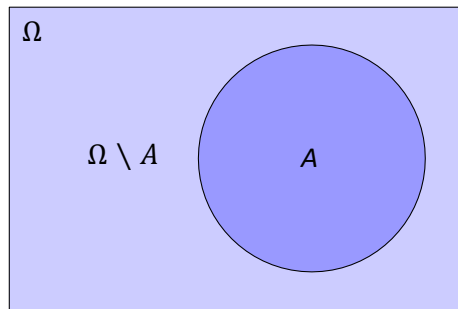
$$A = \{1, 2\},$$

$$P(A) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

$$\overline{A} =$$

More Consequences: Negation Rule

$$P(\bar{A}) = P(\Omega \setminus A) = 1 - P(A)$$

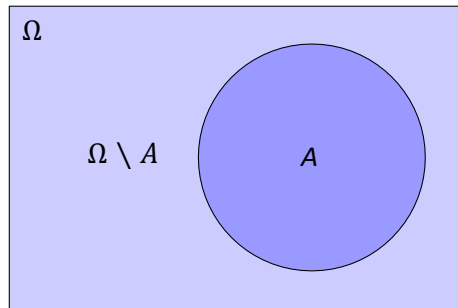


Example:

$$\begin{aligned} A &= \{1, 2\}, & P(A) &= \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \\ \bar{A} &= \{3, 4, 5, 6\}, & P(\bar{A}) &= \end{aligned}$$

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Example:

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Random (Stochastic) Variables

- Outcomes in a sample space need not be numerical, e.g., die rolls, heads/tails, playing cards, words, etc.
- To work with outcomes mathematically, we often map them to numbers.
- A random variable performs this mapping.

Random (Stochastic) Variables

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- To work with outcomes mathematically, we often map them to numbers.
- A random variable performs this mapping.

A *random variable* (*r.v.*) is neither random nor a variable. It's a function that does exactly what we need, e.g.,

$$X: \Omega \rightarrow \mathbb{R}$$

Continuous Distributions: Probability Density Functions

For continuous random variables, the *probability density function* (*PDF*) assigns a non-negative density to each possible value of X .

$$p(x): X \rightarrow \mathbb{R}_{\geq 0}$$

- $p(x)$ describes how probability *density* is distributed over the range of X , e.g., the probability of $0 \leq X \leq 1$ is given by:

$$P[0 \leq X \leq 1] = \int_0^1 p(x) \, dx$$

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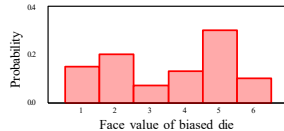
$$P[0 \leq X \leq 1] = \int_0^1 p(x) dx$$

- We say that X is distributed according to $p(x)$, written informally as:

$$X \sim p(x)$$

Types of Random Variables

- The range of a **discrete random variable** is a countable set, e.g., $\{1, 2, \dots, 6\}$.

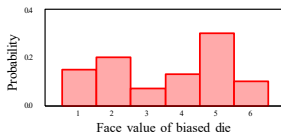


discrete pmf

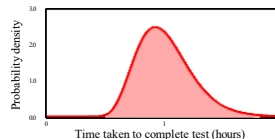
Figure taken from Computer Vision: models, learning and inference by Simon Prince.

Types of Random Variables

- The range of a **discrete random variable** is a countable set, e.g., $\{1, 2, \dots, 6\}$.
- The range of a **continuous random variable** is an uncountable set, e.g., the reals $\mathbb{R}$, or $[0, 1]$.



discrete pmf



continuous pdf

Figures taken from Computer Vision: models, learning and inference by Simon Prince.

PMF vs PDF: Examples



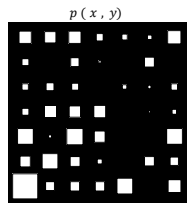
- Discrete events: either 1, 2, 3, 4, 5, or 6.
- Probability **mass** function (PMF)
 $P[X = d] = p(d)$
- $P[X = d] = 1/6, d \in \{1, 2, 3, 4, 5, 6\}$ (fair die)



- Any real number (theoretically infinite)
- Probability **density** function (PDF) $p(x)$
- $P[t_1 < X < t_2]$ involves integrating $p(x)$
- $P[X = t_1] = 0$ but often $p(t_1) \neq 0$

Joint Probabilities

- Consider two random variables X and Y .
- Observe paired instances, (x_k, y_k) , $k = 1, \dots, N$. Some paired outcomes will occur more frequently.
- This information is encoded in the *joint* probability distribution $p(x, y)$ which describes how X and Y vary together.



→ **discrete joint distribution** (PMF-like representation)

Figure from **Computer Vision: models, learning and inference** by Simon Prince.

Marginalization

The distribution of any single variable can be recovered from a joint distribution by summing for the discrete case

$$p(x) = \sum_y p(x, y)$$

and integrating for the continuous case

$$p(x) = \int p(x, y) dy$$

We add up (or integrate over) all ways X could occur together with Y .

Marginalization (cont.)

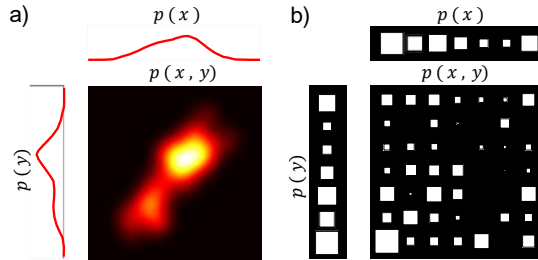


Figure from **Computer Vision: models, learning and inference** by Simon Prince.

Conditional Probabilities

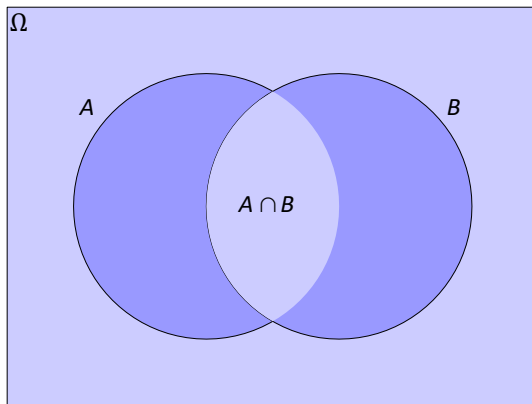
$$P(A|B)$$

The probability of event A *given* that event B has happened

Conditional Probabilities

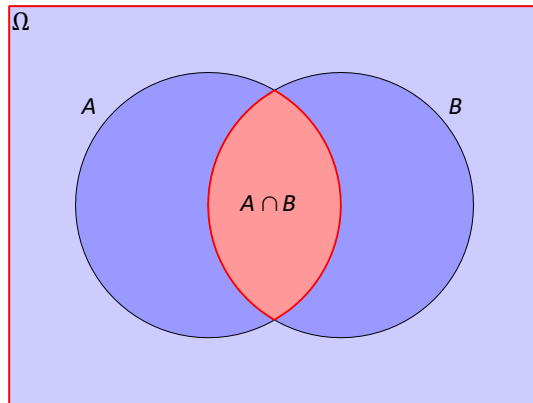
This is not the probability that event A **and** B will happen

$$P(A|B) \neq P(A \cap B)$$



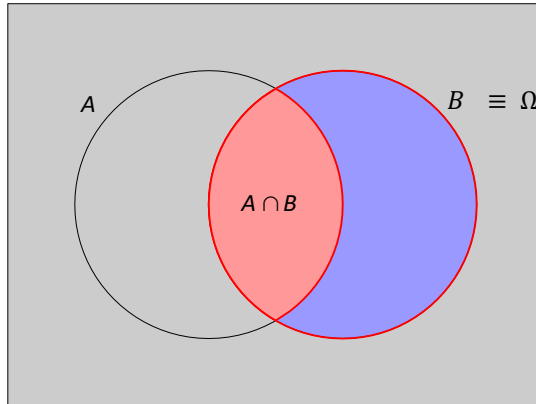
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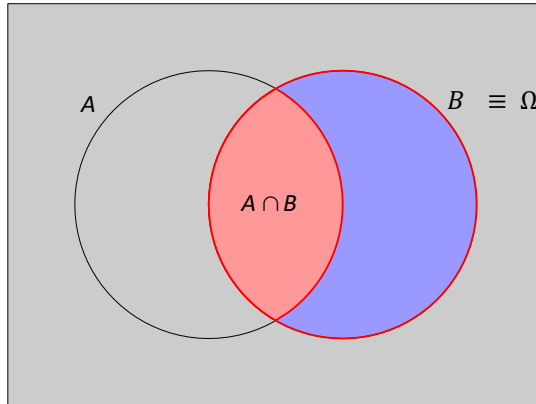
Conditional Probabilities

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Conditional Probabilities

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



Conditional Probability (Random Variables)

- The *conditional probability distribution* of X , given that Y takes value y , tells us how likely different values of x are, assuming $Y = y$.
- The conditional probability can be found from the joint distribution function $p(x, y)$:

$$p(x | y) = \frac{p(x, y)}{p(y)} = \frac{p(x, y)}{\int p(x, y) dx}$$

- Extract an appropriate slice and then normalize it.

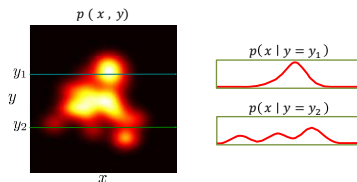


Figure from **Computer Vision: models, learning and inference** by Simon Prince.

Independence

- Two events are *independent* if the joint distribution can be factorized: $P(A \cap B) = P(A)P(B)$
- this means that:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

Independence

- Two events (or random variables) are *independent* if the joint distribution can be factorized: $P(A \cap B) = P(A)P(B)$
- this means that:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$$

Knowing that B happened tells us nothing new about A happening

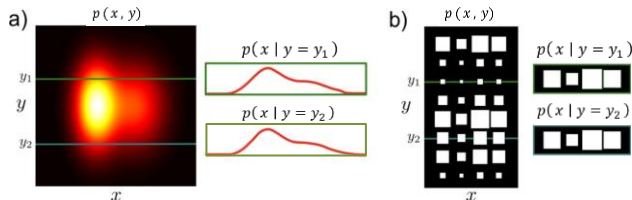


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Bayes' Theorem

if

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

then

$$P(A \cap B) = P(A | B) P(B)$$

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then

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$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

Memorize this and its derivation!

Monty Hall Solution with Bayes



Contestant chose Door 1. C_i : car behind Door i . M_2 : Monty opens Door 2.

Which door should we choose?

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

Priors: $P(C_1), P(C_2), P(C_3)$

Likelihoods: $P(M_2 | C_1), P(M_2 | C_2), P(M_2 | C_3)$

Monty Hall Solution with Bayes

$$P(C_1|M_2) = \frac{P(M_2|C_1) P(C_1)}{P(M_2)}$$



$$P(C_3|M_2) = \frac{P(M_2|C_3) P(C_3)}{P(M_2)}$$

Contestant chose Door 1. C_i : car behind Door i . M_2 : Monty opens Door 2.

Priors: $P(C_1) = P(C_2) = P(C_3) = 1/3$

Monty Hall Solution with Bayes

$$P(C_1|M_2) = \frac{P(M_2|C_1) P(C_1)}{P(M_2)}$$



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Contestant chose Door 1. C_i : car behind Door i . M_2 : Monty opens Door 2.

Priors: $P(C_1) = P(C_2) = P(C_3) = 1/3$

Likelihoods: $P(M_2 | C_1) = 1/2, P(M_2 | C_2) = 0, P(M_2 | C_3) = 1$.

Monty's behavior is *not random*, that's why the likelihoods differ.

Monty Hall Solution with Bayes

$$P(C_1|M_2) = \frac{P(M_2|C_1) P(C_1)}{P(M_2)}$$



$$P(C_3|M_2) = \frac{P(M_2|C_3) P(C_3)}{P(M_2)}$$

Contestant chose Door 1. C_i : car behind Door i . M_2 : Monty opens Door 2.

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Likelihoods: $P(M_2 | C_1) = 1/2, P(M_2 | C_2) = 0, P(M_2 | C_3) = 1$.

$$P(M_2) = P(M_2|C_1)P(C_1) + P(M_2|C_2)P(C_2) + P(M_2|C_3)P(C_3) = 1/2$$

Monty Hall Solution with Bayes

$$P(C_1|M_2) = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3}$$



$$P(C_3|M_2) = \frac{1 \times \frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$$

Contestant chose Door 1. C_i : car behind Door i . M_2 : Monty opens Door 2.

Priors: $P(C_1) = P(C_2) = P(C_3) = 1/3$

Likelihoods: $P(M_2 | C_1) = 1/2, P(M_2 | C_2) = 0, P(M_2 | C_3) = 1$.

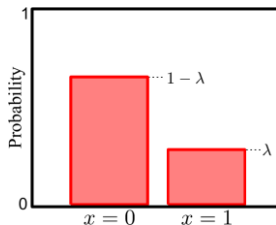
$$P(M_2) = P(M_2|C_1)P(C_1) + P(M_2|C_2)P(C_2) + P(M_2|C_3)P(C_3) = 1/2$$

Bernoulli: binary variables

- Domain: binary variables ($x \in \{0, 1\}$)
- Parameters: $\lambda = P[x = 1], \lambda \in [0, 1]$

Then $P[x = 0] = 1 - \lambda$, and

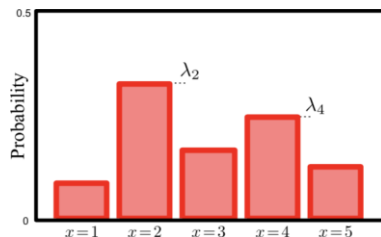
$$p(x) = \lambda^x(1 - \lambda)^{1-x} = \begin{cases} \lambda, & \text{if } x = 1 \\ 1 - \lambda, & \text{if } x = 0 \end{cases}$$



Categorical

- Domain: discrete variables ($X \in \{x_1, \dots, x_K\}$)
- Parameters: $\lambda = [\lambda_1, \dots, \lambda_K]$
- with $\lambda_k \in [0, 1]$ and $\sum_{k=1}^K \lambda_k = 1$

$$P[X = x_k] = \lambda_k$$

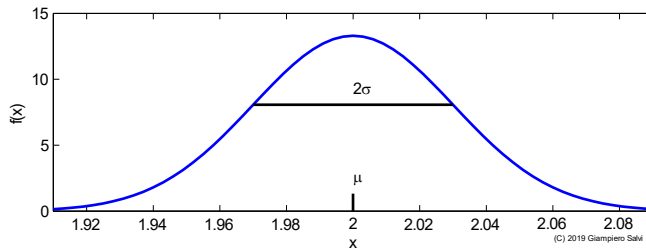


Gaussian distributions: One-dimensional

- Univariate normal distribution
- Domain: real numbers ($x \in \mathbb{R}$)

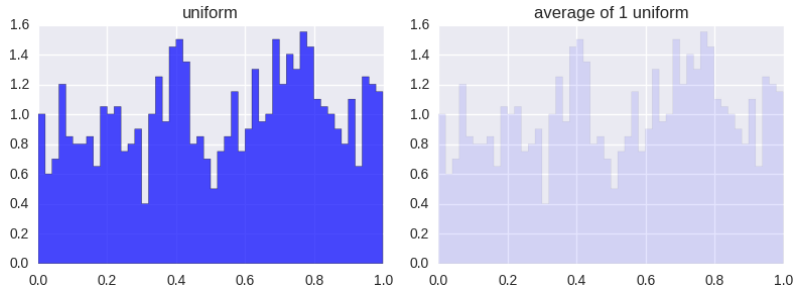
$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right]$$



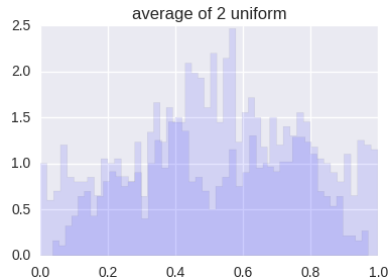
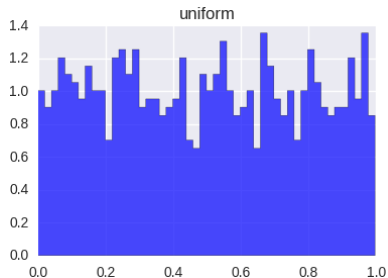
Why Gaussian: Central Limit Theorem

For a large number of independent, identically distributed (iid) variables with finite mean and variance, the distribution of their sum or average is approximately Gaussian, even when the individual variables are not.



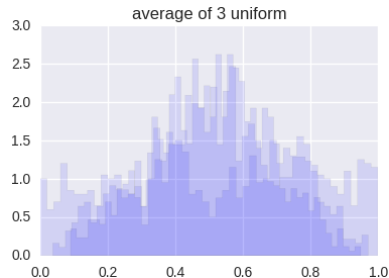
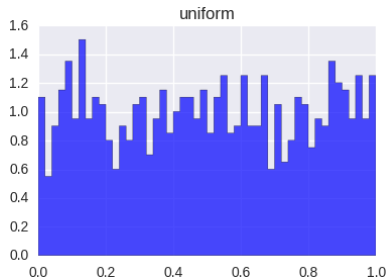
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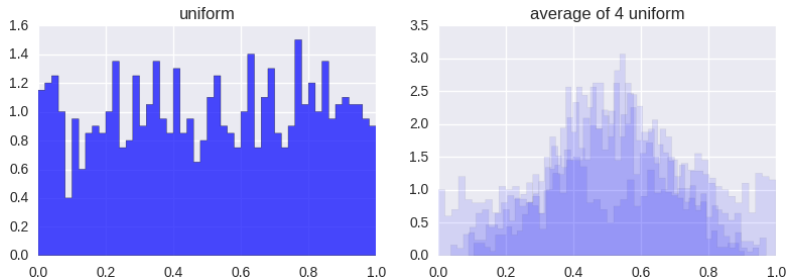
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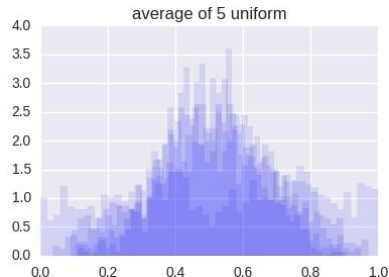
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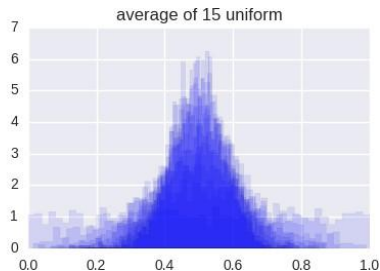
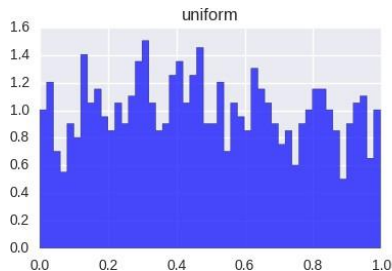
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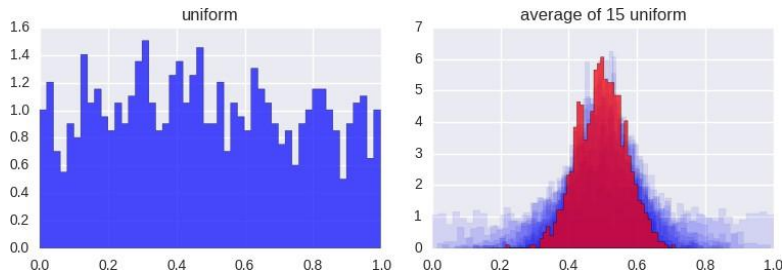
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Averaging many noisy measurements makes the result more predictable.

Why Gaussian: Central Limit Theorem

Galton Board (Sir Francis Galton, 1822-1911)



Expected value

$$\mathbb{E}[\mathbf{X}] = \mu = \int x \, p(x) \, dx$$

- The “center of gravity” of a distribution

Expected value

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- The “center of gravity” of a distribution
- Sampled expected value (mean)

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \approx \mu$$

Variance

$$\text{Var}(X) = \sigma^2 = \mathbb{E}[(X - \mu)^2]$$

- The “spread” of a distribution

Variance

$$\text{Var}(X) = \sigma^2 = \mathbb{E}[(X - \mu)^2]$$

- The “spread” of a distribution
- Sampled variance

$$\widehat{\sigma^2} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$$

Covariance

Scalar covariance (two variables)

$$\sigma_{X,Y} = \mathbb{E}[(X - \mu_x)(Y - \mu_y)]$$

Vector generalization (covariance matrix)

$$\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T], \quad \boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$$

- Shows how two variables vary *together*

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- Shows how two variables vary *together*
- Sample covariance

$$\widehat{\sigma_{X,Y}} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})$$

Gaussian distributions: D Dimensions

- Also known as the *multivariate* normal distribution
- Domain: real numbers ($\mathbf{x} \in \mathbb{R}^D$)

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_D \end{bmatrix}$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \dots \\ \mu_D \end{bmatrix}$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1D} \\ \sigma_{21} & \sigma_2^2 & \dots & \\ \dots & & & \\ \sigma_{D1} & \dots & & \sigma_D^2 \end{bmatrix}$$

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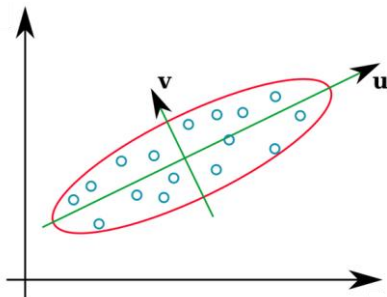
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$$p(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{D}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right]$$

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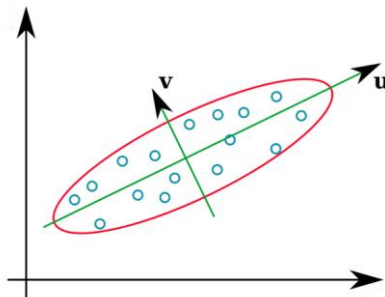
Eigenvalue decomposition of the covariance matrix: $\boldsymbol{\Sigma} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^T$



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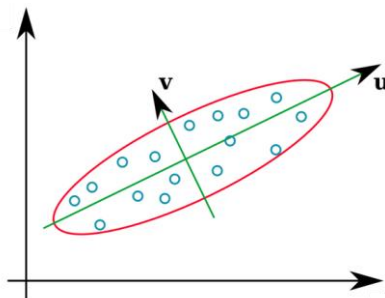


Eigenvectors = directions
of principal variance

Gaussian distributions

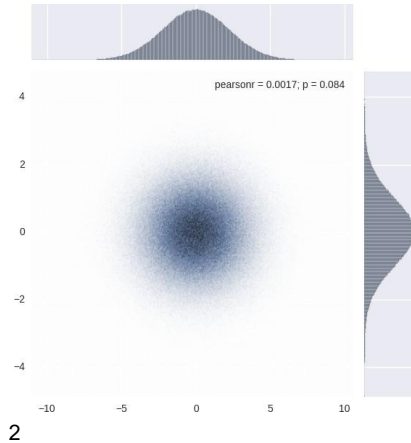
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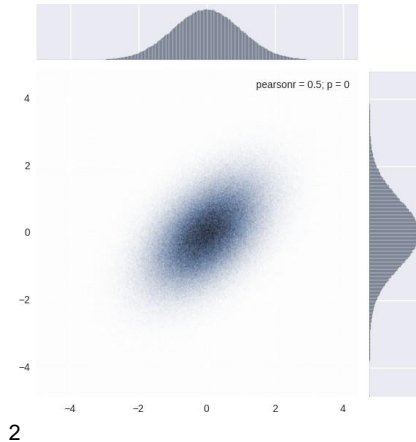
Eigenvalues = variance
along those directions

Examples



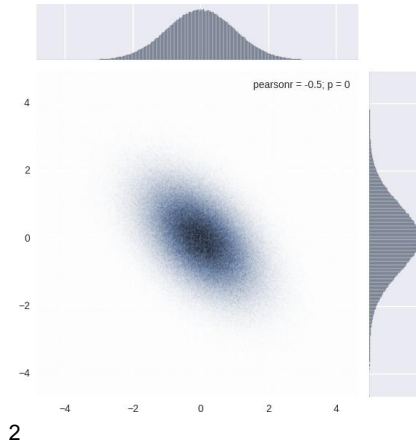
²Script by C.E. Ek

Examples



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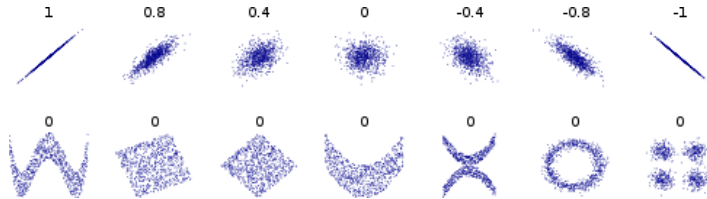
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²Script by C.E. Ek

Covariance and Independence

- covariance captures linear association
- variables may have zero covariance but still be dependent

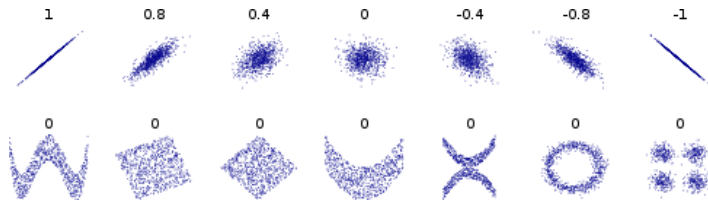


3

³Figure adapted from Wikipedia

Covariance and Independence

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- Can anyone remember an important case where zero covariance **does** imply independence?



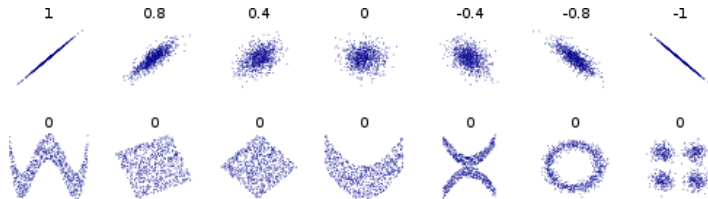
3

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Covariance and Independence

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- Can anyone remember an important case where zero covariance **does** imply independence?

For **jointly Gaussian random variables**, zero covariance implies independence. That is a special property of the Gaussian family.



3

³Figure adapted from Wikipedia

Outline

1 Probability Theory Reminder

- Axioms and Properties
- Distributions and Moments

2 Probabilistic Machine Learning

- Supervised Learning, General Definition
- Regression
- Classification

General ML problem (supervised learning)

Data:

$$\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$$

Where $\mathbf{x}$ denotes the input features, and y the target (output)

Task type:

- if y is discrete: *classification*
- if y is continuous: *regression*

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Task type:

- if y is discrete: *classification*
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Learning: understand how variables are related

- estimate the parameters of a probabilistic model from observations $\{(\mathbf{x}_j, y_j)\}$

Inference: predict an answer given an observation

- compute the conditional distribution $p(y|\mathbf{x})$

Inference (Bayes' Rule)

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)}$$

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$$p(x) = \begin{cases} \sum_y p(x|y) p(y) & \text{classification} \\ \int p(x|y) p(y) dy & \text{regression} \end{cases}$$

Probabilistic Regression

Regression via conditional probability

- Model the joint distribution $p(x, y)$
- Obtain the conditional distribution

$$p(y|x) = \frac{p(x, y)}{p(x)}$$

- Compute conditional expectation $\mathbb{E}[Y | X = x]$

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Direct probabilistic regression model

- Define $y = f(x) + \epsilon$
- Specify a distribution for the noise $\epsilon \sim p(\epsilon)$
- Learn the parameters of $f(x)$ from data

Probabilistic Linear Regression Model

Model:

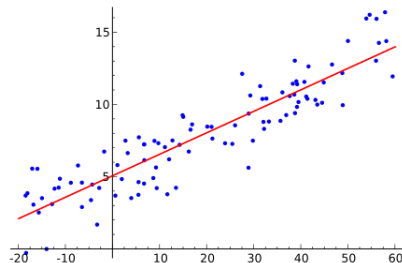
$$y = \mathbf{w}^T \mathbf{x} + \epsilon$$

Assume Gaussian noise:

$$\epsilon \sim \mathcal{N}(0, \sigma^2)$$

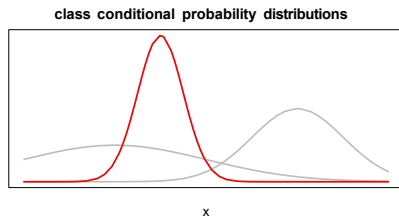
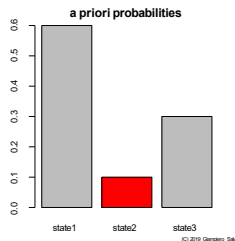
Therefore:

$$Y|X = x \sim \mathcal{N}(\mathbf{w}^T \mathbf{x}, \sigma^2)$$



The Probabilistic Model of Classification

- One of K states is selected with *a priori* probability
 $p(Y = y)$
- Given state y , x is drawn from $p(x|Y = y)$



(C) 2019 Gianpiero Salvi

Problem

For an observed x , we know $p(Y = y)$ and $p(x | Y = y)$ for each of K classes. What is the posterior $p(Y = y | x)$?

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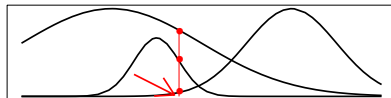
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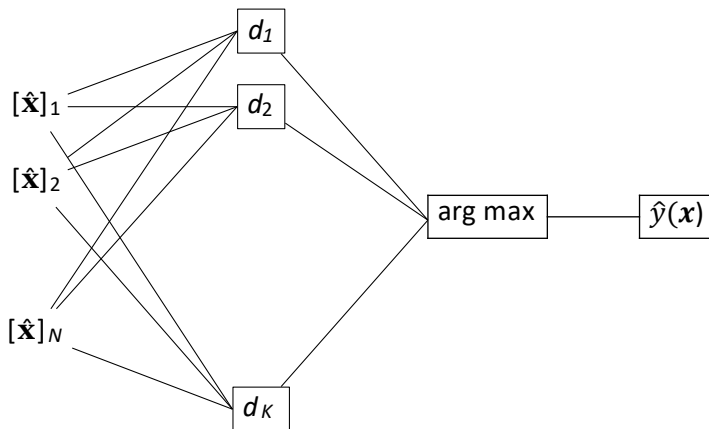
$$\begin{aligned} p(x) &= p(x | Y = y_1) p(Y = y_1) \\ &\quad + p(x | Y = y_2) p(Y = y_2) \\ &\quad + p(x | Y = y_3) p(Y = y_3) \end{aligned}$$



x

Classifiers: Discriminant Functions

$$d_k(\bar{x}) = p(\bar{x} \mid Y = y_k) p(Y = y_k)$$



Example: Is this online product review real or fake?

Task: Determine whether an online product review is real or fake given its length in characters.

The image shows the Fakespot Analyzer website. At the top, it says "Say no to fake reviews and counterfeits" and "Fakespot analyzes reviews to help you make better purchasing decisions". Below this is the "FAKESPOT Analyzer" logo. There is a text input field with the placeholder "Paste an URL you would like to analyze here" and a link "Show me how Fakespot works". The input field contains the text "https://www.". Below the input field is a blue button labeled "Analyze Reviews" with a magnifying glass icon. At the bottom, it says "You can currently analyze links from" and lists logos for "e-Commerce" (amazon, BEST BUY, SEPHORA, STEAM, Walmart) and "Hospitality" (tripadvisor, yelp).

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Task: Determine whether an online product review is real or fake given its length in characters.

Notation:

- Let Ω be the set of all possible online product reviews.
- Let $Y: \Omega \rightarrow \{+1, -1\}$ be the class label random variable: +1 for Real, -1 for Fake.
- Let $X: \Omega \rightarrow \mathbb{N}$ be the review length in characters, with observed value x .

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Information given:

- We have a dataset of reviews labeled as real or fake.
- From this dataset, we estimate the class priors:
 $p(Y = +1) = 0.25$, $p(Y = -1) = 0.75$
- We also estimate $p(x \mid Y = +1)$ and $p(x \mid Y = -1)$,
after choosing an appropriate model for review length.

Example: Is this online product review real or fake?

Task: Given review length x , compute the probability that the review is real: $p(Y = +1 | x)$.

Solution: Apply Bayes' Rule!

$$\begin{aligned} p(Y = +1 | x) &= \frac{p(x | Y = +1) p(Y = +1)}{p(x)} \\ &= \frac{p(x | Y = +1) p(Y = +1)}{p(x | Y = +1) p(Y = +1) + p(x | Y = -1) p(Y = -1)} \end{aligned}$$

The evidence $p(x)$ is the marginal density, found by summing over both classes. It ensures the posteriors sum to one.

MAP Classification

- **Maximum A Posteriori (MAP):**

Choose the class y with the highest posterior probability given x .

$$\begin{aligned}\hat{y}_{MAP} &= \arg \max_y p(y | x) \\ &= \arg \max_y \frac{p(x | y) p(y)}{p(x)} \\ &= \arg \max_y p(x | y) p(y)\end{aligned}$$

$p(x)$ is the same for every class, so it does not affect the argmax.

Here $p(y | x)$, $p(x | y)$ and $p(y)$ abbreviate notation with $Y = y$.

Maximum-Likelihood Classification Rule

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- **Maximum Likelihood (ML):**

With equal class priors, $p(y)$ is constant across classes, so MAP reduces to maximum-likelihood classification.

$$\hat{y}_{ML} = \arg \max_y p(x | y)$$

Next lecture: maximum-likelihood parameter estimation fits model parameters to training data. Here, we choose a class for an observed x .

Summary

Today:

- 1 Probability Theory Reminder
 - Axioms and Properties
 - Distributions and Moments

- 2 Probabilistic Machine Learning
 - Supervised Learning, General
 - Definition Regression
 - Classification

Next time: How to *fit* probability models to data (estimating parameters with the maximum likelihood optimality criterion)